Tampines Regional Library
LearnX Community
Raspberry Pi Python Workshop
Course Material
By
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PLEASE DO NOT WRITE ANYTHING ON THE PAGES OF THIS BOOKLET. THANK YOU.

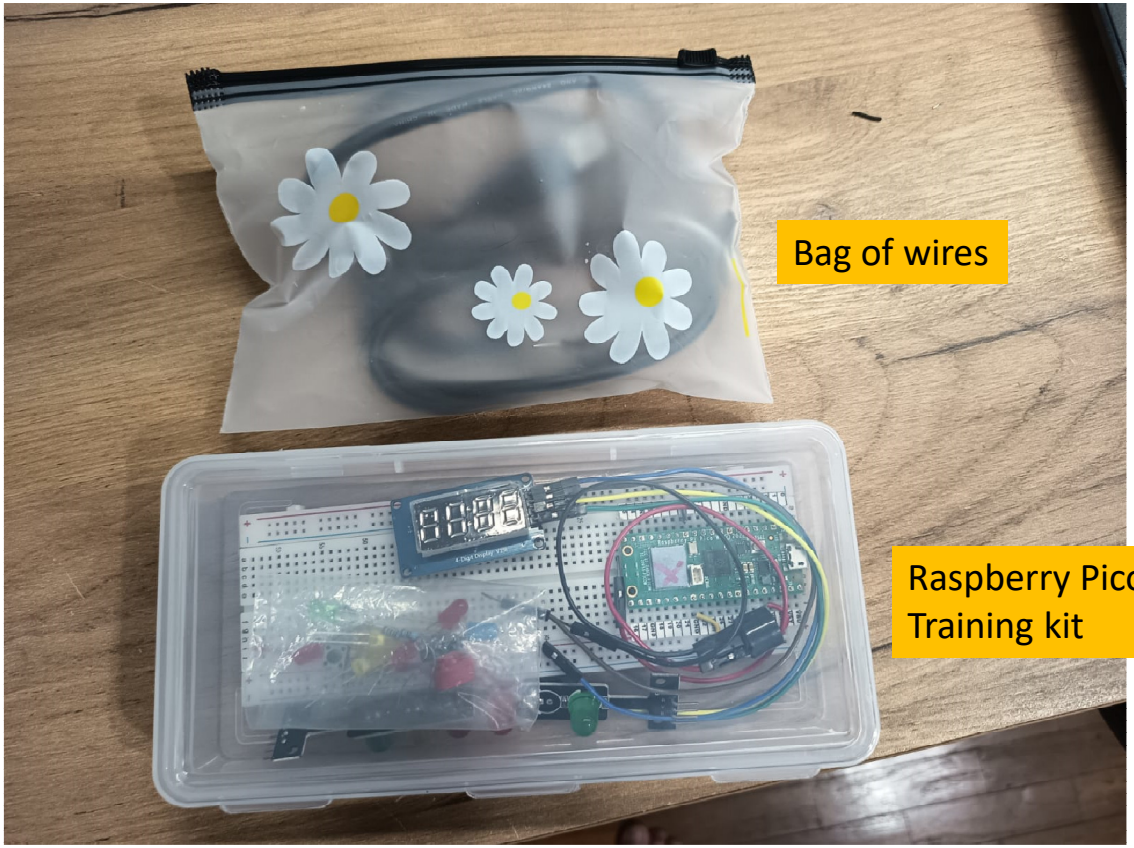
Raspberry Pi Python Workshop

Step into the world of physical computing with our **Raspberry Pi Python Workshop**. This hands-on experience transforms abstract logic into tangible action as you program the **Raspberry Pi Pico**, watching your code come to life the moment it is executed.

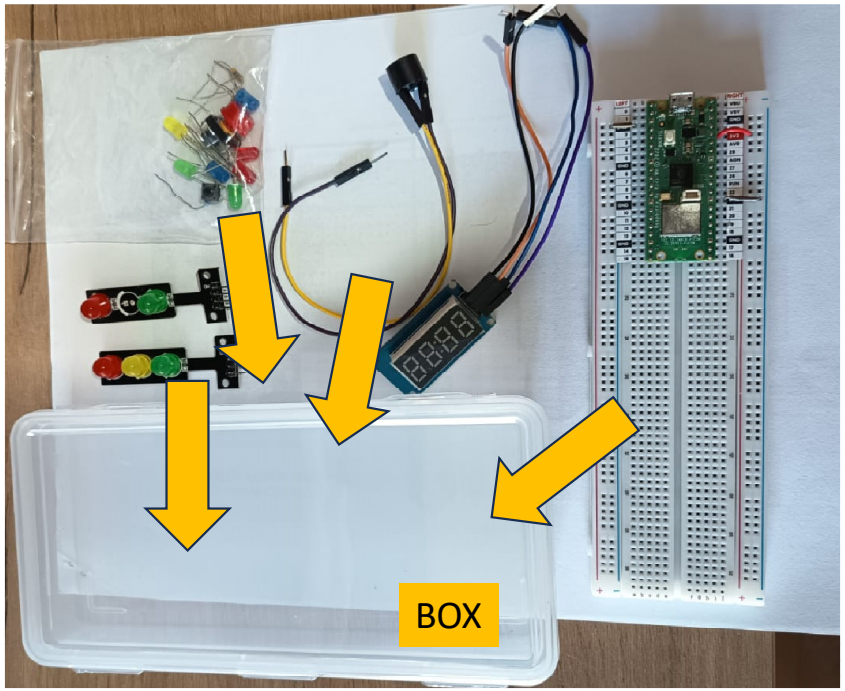
Master the following core pillars of Python through practical application:

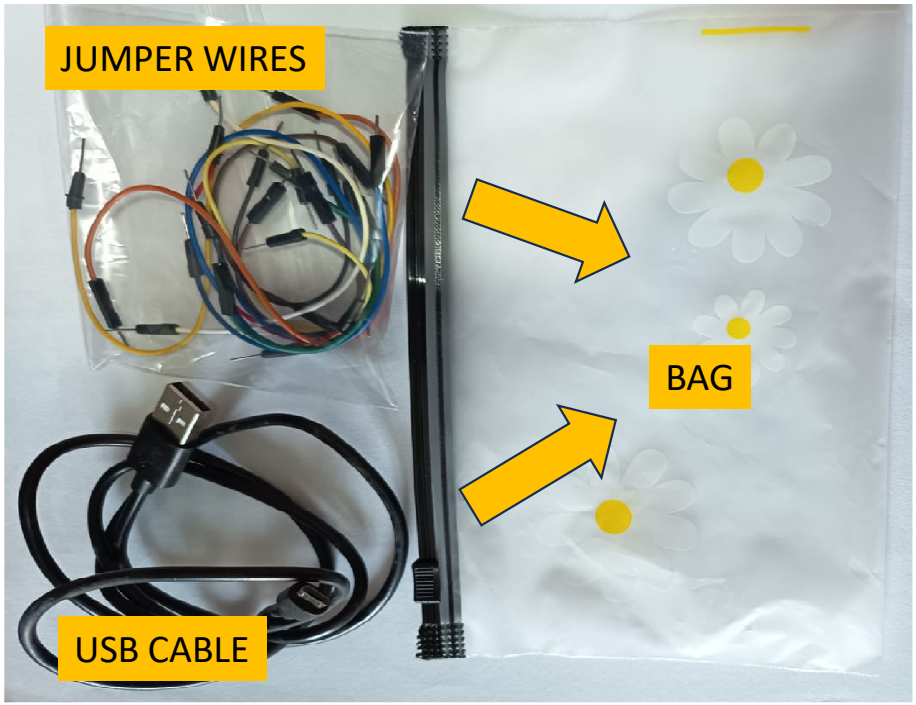
- **The Thonny IDE:** Your command center for writing, debugging, and uploading code to the Pico.
- **Python Syntax & Rules:** The grammar of programming that ensures your instructions are understood.
- **Variables:** Containers used to store and label data for later use.
- **Functions:** Reusable blocks of code designed to perform specific tasks efficiently.
- **If/Else Logic:** The "brain" of your program, allowing it to make decisions based on different conditions.
- **While & For Loops:** Mechanisms to automate repetitive tasks or keep programs running indefinitely.
- **Python Libraries:** Pre-written toolsets that extend the Pico's capabilities without writing code from scratch.
- **Error Messages:** Learning to read the "language of mistakes" to troubleshoot and refine your projects.
- **AI Integration:** Strategies for using Artificial Intelligence as a personal tutor to accelerate your ongoing learning journey.

Please Return them like this - tomorrow



THE RASPBERRY PI PICO TRAINING KIT





PLEASE PACK THE KIT AT THE END OF THIS WORKSHOP INTO IT'S ORIGINAL PACKAGING. THANK YOU

Python Lesson – Session # 1

LIBRARIES
- machine
- time
- picozero

VARIABLES/NAMES

LOOPS
- while True:

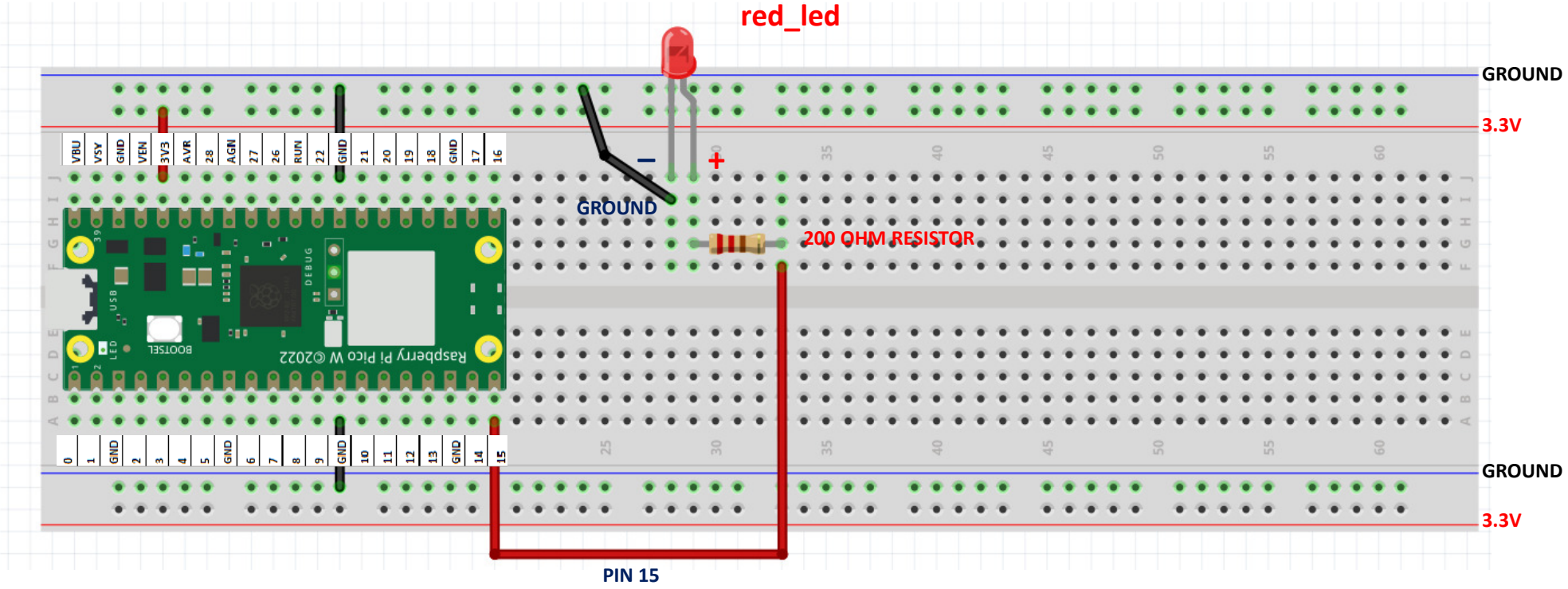
PYTHON RULE
- Indentation / Code Blocks
- Character casing

SAVING YOUR WORK
- All Python files has .py extension

ALGORITHM
- Formulating what you want the program to do / Recipe

EXERCISE #1 Wiring a Light Emitting Diode (LED)

Please wire up the following circuit



TESTING OUR CIRCUIT USING THONNY SHELL

Ex 1a. Turning our Red LED on

```
>>> from picozero import LED
>>> red_led = LED(15)
```

Library

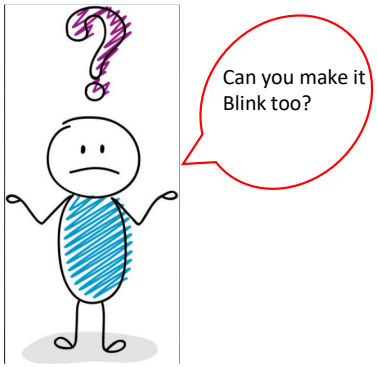
Variable

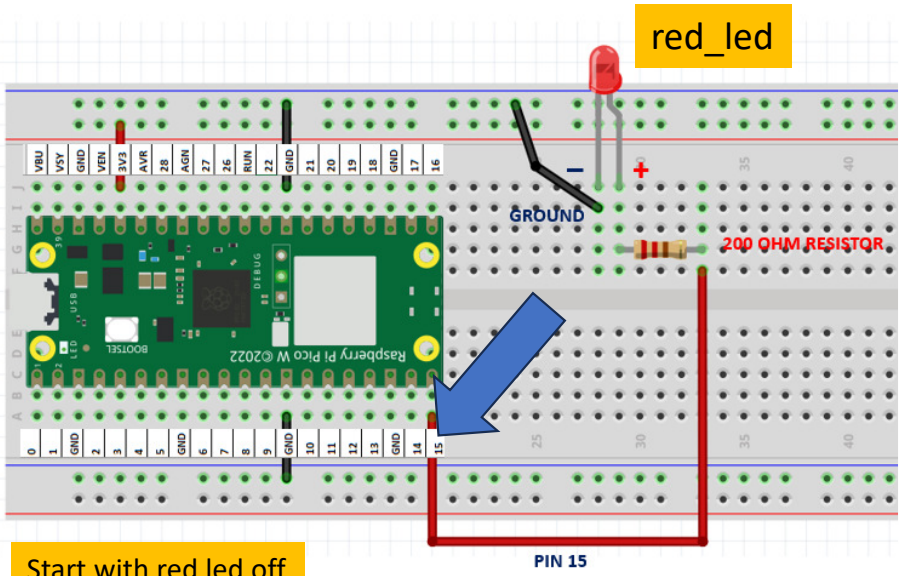
```
>>> red_led.on()
```

Ex 1b. Turning it off

```
>>> red_led.off()
```

Ex 1c. Blinking the red LED

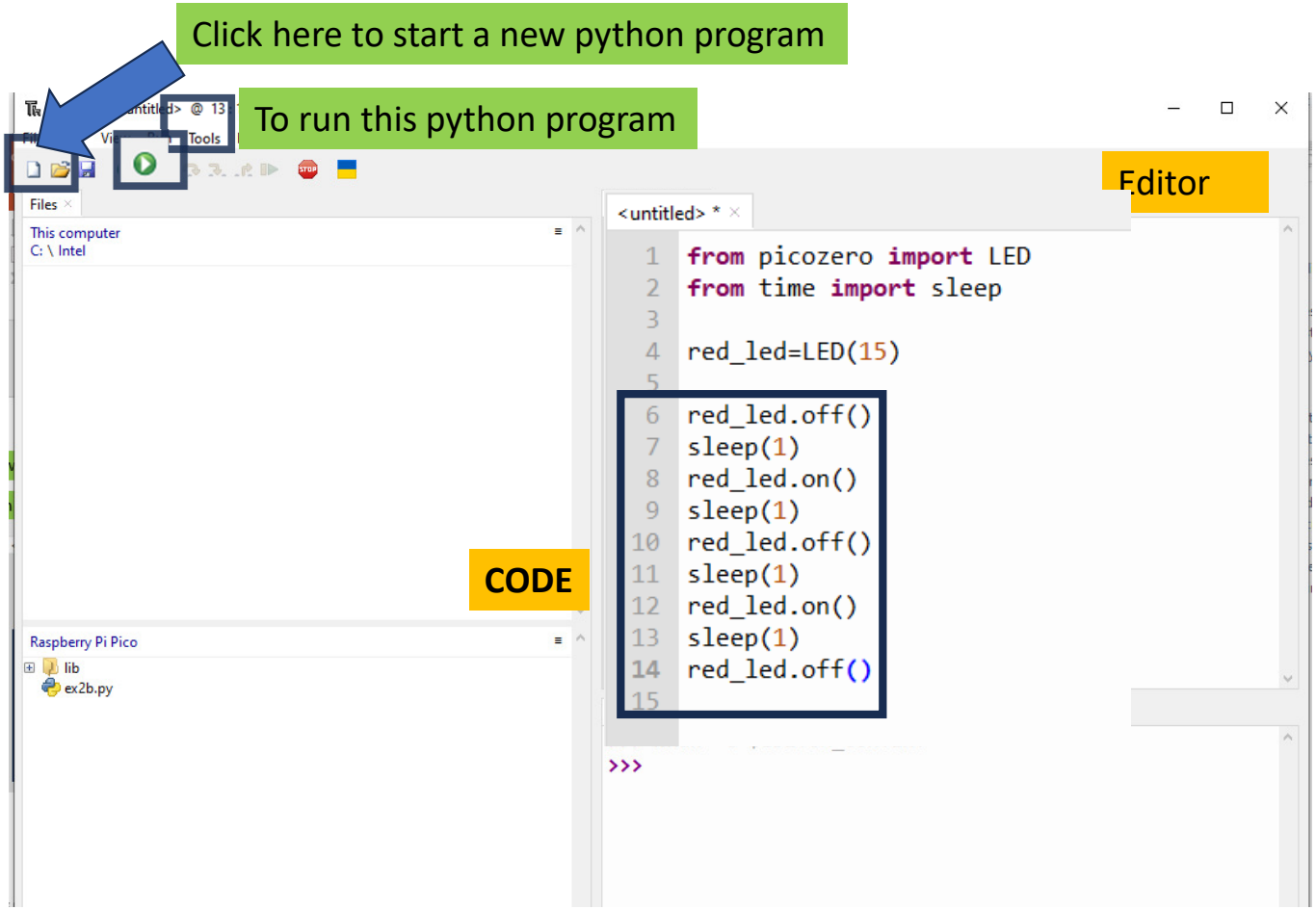




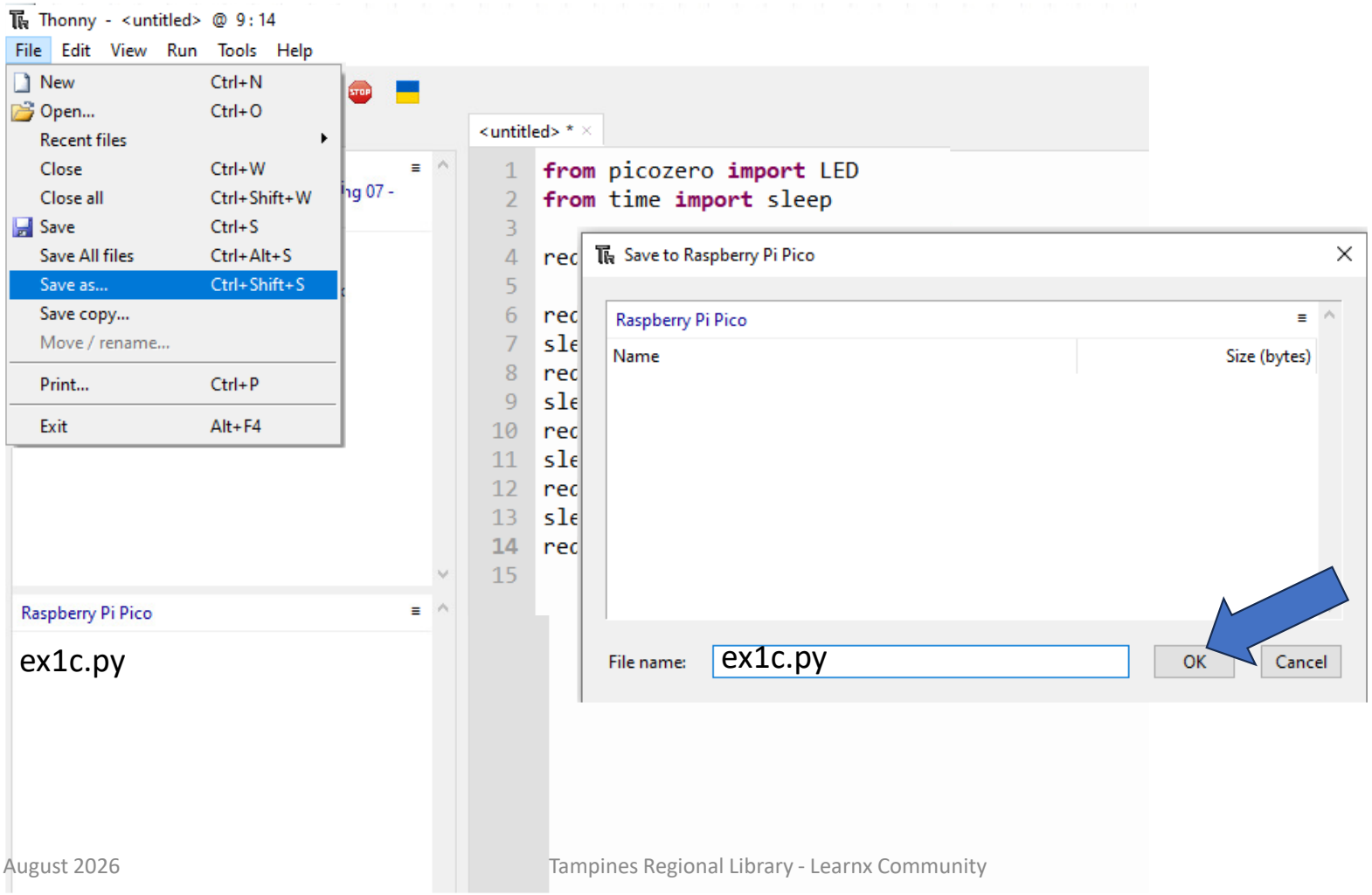
Start with red led off
Wait 1 second
Turn red led on
Wait 1 second
Turn red led off
Wait 1 second
Turn red led on
Wait 1 second
Turn red led off

ALGORITHM TO CODE

Start with red led off
Wait 1 second
Turn red led on
Wait 1 second
Turn red led off
Wait 1 second
Turn red led on
Wait 1 second
Turn red led off



SAVING A PROGRAM



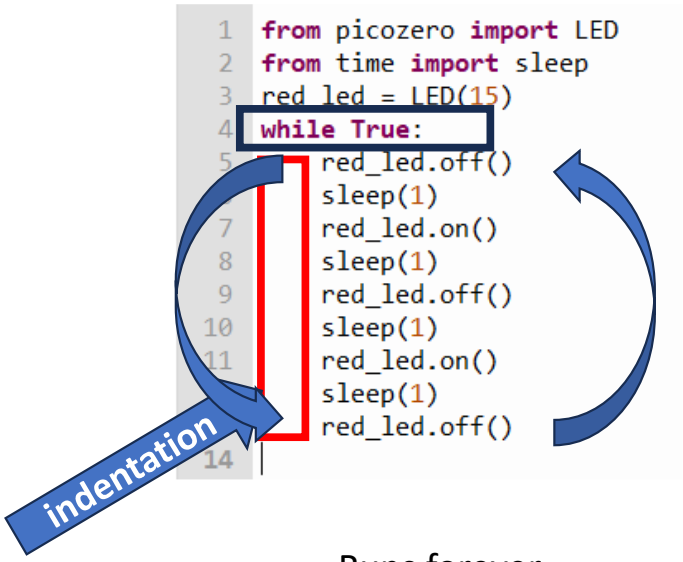
ex1c.py

```
1  from picozero import LED
2  from time import sleep
3  red_led = LED(15)
4
5  red_led.off()
6  sleep(1)
7  red_led.on()
8  sleep(1)
9  red_led.off()
10 sleep(1)
11 red_led.on()
12 sleep(1)
13 red_led.off()
```

Runs once

Convert

Python Syntax Rules

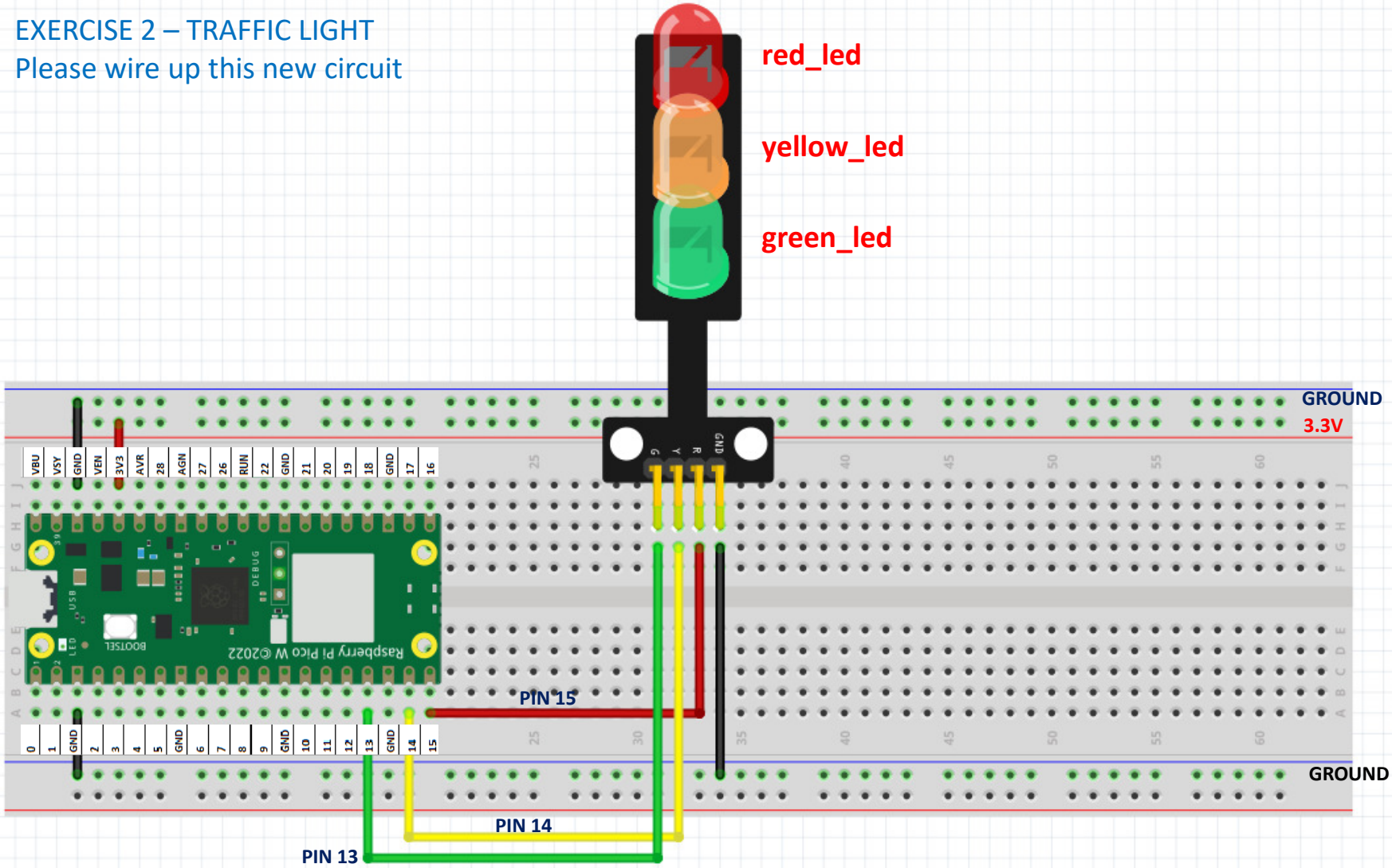


Runs forever

Save it as ex1d.py
Run this program

EXERCISE 2 – TRAFFIC LIGHT

Please wire up this new circuit

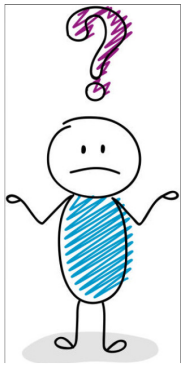


EXERCISE 2 – CONTROLLING MULTIPLE LEDs

Ex 2a. Test the Circuit. Turn each of the LEDs ON and OFF

```
>>> from picozero import LED
>>> red_led= LED(15)
>>> yellow_led = LED(14)
>>> green_led=LED(13)
```

Ex 2b. Turning the circuit into a Traffic Light System



How?

How does a traffic light work?

Green Light On
Give cars 10 seconds to cross
Green Light Off
Yellow Light On
Give 5 seconds warning
Yellow Light Off
Red Light On
Wait 10 seconds
Red Light Off

Algorithm

Code

```
from picozero import LED
from time import sleep
red_led = LED(15)

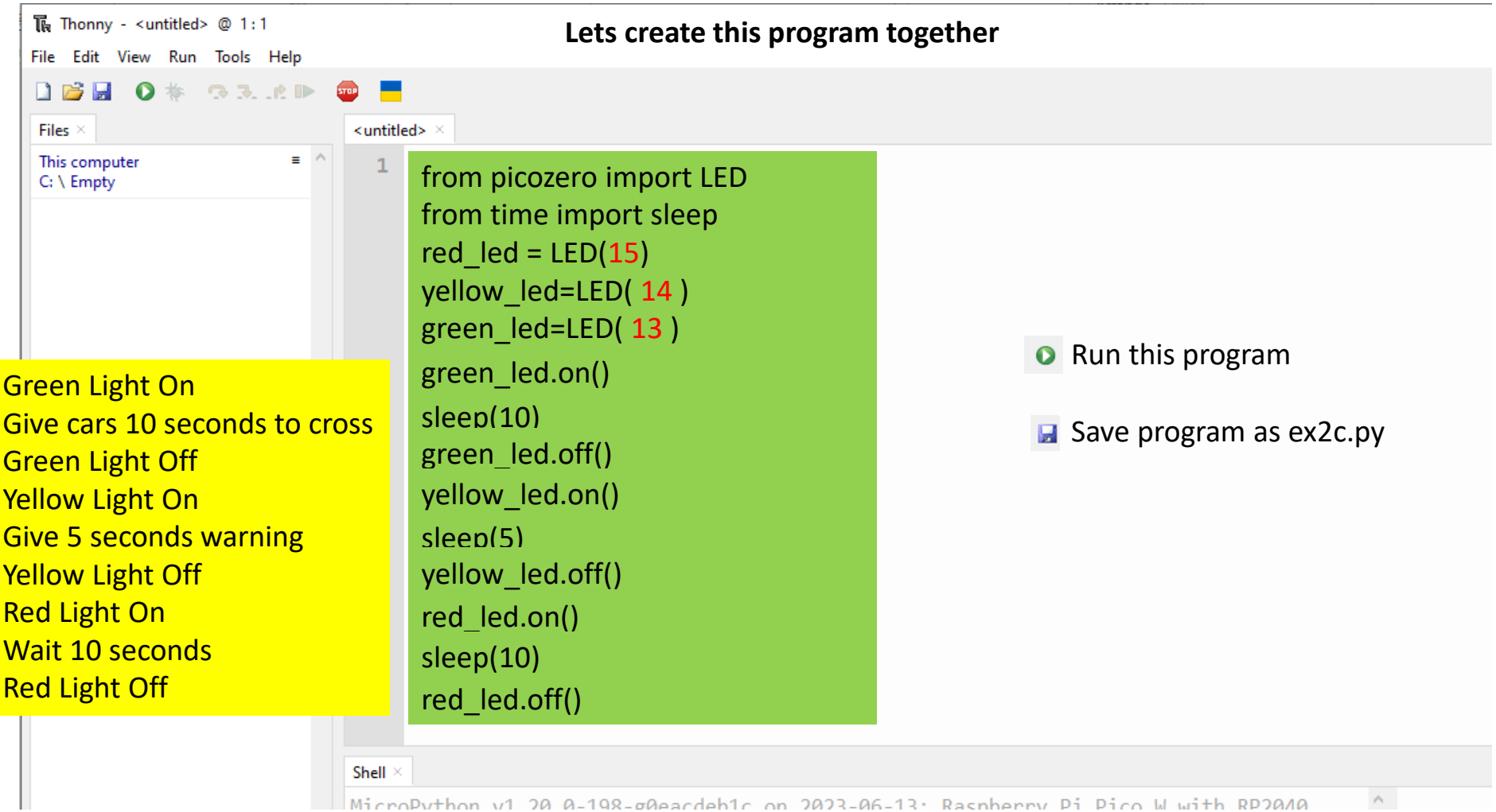
red_led.off()
sleep(1)
red_led.on()
sleep(1)
red_led.off()
sleep(1)
red_led.on()
sleep(1)
red_led.off()
```

hint

```
green_led.on()
sleep(10)
green_led.off()
yellow_led.on()
sleep(5)
yellow_led.off()
red_led.on()
sleep(10)
red_led.off()
```

Code this in the editor
Run and save it as ex2b.py

Lets create this program together



Green Light On
Give cars 10 seconds to cross
Green Light Off
Yellow Light On
Give 5 seconds warning
Yellow Light Off
Red Light On
Wait 10 seconds
Red Light Off

Run this program

Save program as ex2c.py

ex2c.py

```
1  from picozero import LED
2  from time import sleep
3  red_led = LED(15)
4  yellow_led=LED(14)
5  green_led=LED(13)
6
7  green_led.on()
8  sleep(10)
9  green_led.off()
10 yellow_led.on()
11 sleep(5)
12 yellow_led.off()
13 red_led.on()
14 sleep(10)
15 red_led.off()
```

Runs once

convert

Runs forever

```
1  from picozero import LED
2  from time import sleep
3  red_led = LED(15)
4  yellow_led=LED(14)
5  green_led=LED(13)
6  while True:
7      green_led.on()
8      sleep(10)
9      green_led.off()
10     yellow_led.on()
11     sleep(5)
12     yellow_led.off()
13     red_led.on()
14     sleep(10)
15     red_led.off()
16
```

Run the Program
Save it as ex2d.py

Python Lesson – Session # 2

FUNCTIONS

LOOPS

- for loop

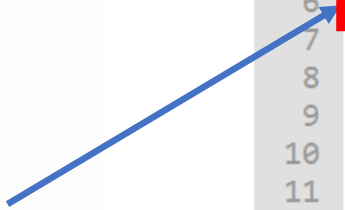
MESSAGING

- how humans and machine communicate with each other

FUNCTION – A SET OF CODES WITH A NAME

ex2d.py

```
1  from picozero import LED
2  from time import sleep
3  red_led = LED(15)
4  yellow_led=LED(14)
5  green_led=LED(13)
6  while True:
7      green_led.on()
8      sleep(10)
9      green_led.off()
10     yellow_led.on()
11     sleep(5)
12     yellow_led.off()
13     red_led.on()
14     sleep(10)
15     red_led.off()
16
```



```
1  from picozero import LED
2  from time import sleep
3  red_led = LED(15)
4  yellow_led=LED(14)
5  green_led=LED(13)
6  def trafficlight():
7      green_led.on()
8      sleep(10)
9      green_led.off()
10     yellow_led.on()
11     sleep(5)
12     yellow_led.off()
13     red_led.on()
14     sleep(10)
15     red_led.off()
16
17 while True:
18     trafficlight()
19
```

Run the Program
Save it as ex2e.py

BUTTONS ARE EVERYWHERE IN THE REAL WORLD

INTERACTION BETWEEN HUMANS AND THE REAL WORLD VIA A COMPUTER SYSTEM



HUMANS CANNOT DIRECTLY INTERACT WITH ELEVATOR SYSTEM



ELEVATOR CONTROL SYSTEM

HOW DO HUMANS AND MACHINES COMMUNICATE IN THE REAL WORLD

WHAT DO YOU SEE AT A PEDESTRIAN CROSSING?

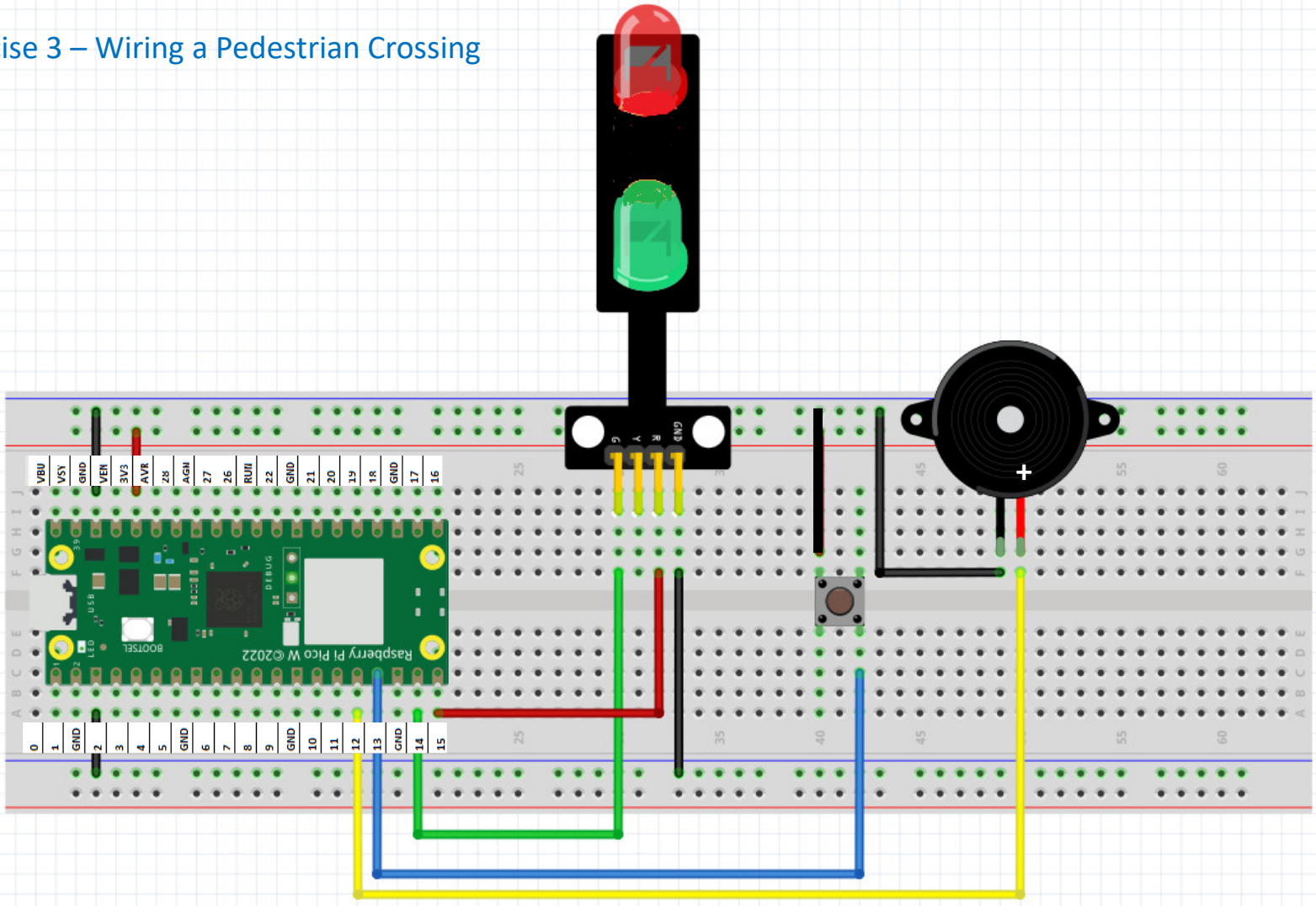


RED AND GREEN MAN
BUTTON
ANTI SPAM LIGHT
BUZZER
BLINKING GREEN MAN
COUNTER



RED AND GREEN MAN	MACHINE TO MAN	STOP OR CROSS
BUTTON	MAN TO MACHINE	I WANT TO CROSS
ANTI SPAM LIGHT	MACHINE TO MAN	SOMEONE ALREADY REQUESTED
BUZZER	MACHINE TO MAN	HURRY
BLINKING GREEN MAN	MACHINE TO MAN	HURRY
COUNTER	MACHINE TO MAN	TIME RUNNING OUT

Exercise 3 – Wiring a Pedestrian Crossing



TESTING OUR CIRCUIT USING THONNY SHELL

Ex 3a. Turning our Buzzer on

```
>>> from picozero import Buzzer
>>> buzzer= Buzzer(12)
>>> buzzer.on()
```

Ex 3b. Turning it off

```
>>> buzzer.off()
```

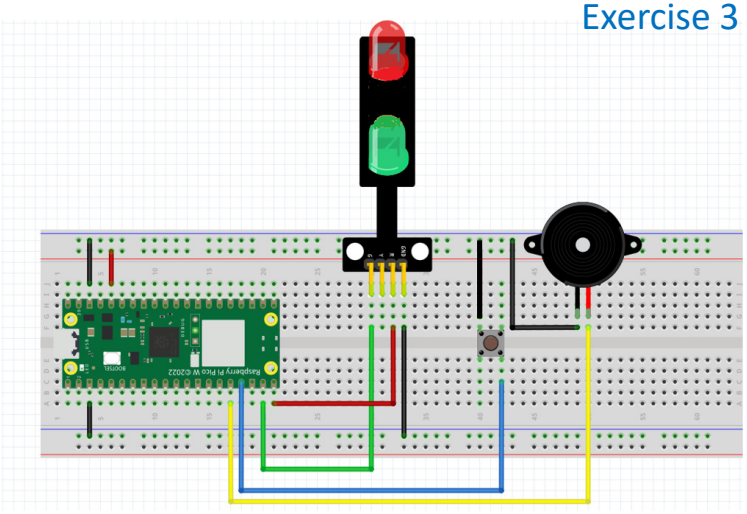
Ex 3b. Setting up multiple sensors : our circuit has Buzzer, Button, LED

```
>>> from picozero import Buzzer, Button, LED
>>> buzzer= Buzzer(12)
>>> button=Button(13)
>>> red_led=LED(15)
>>> button.when_pressed = red_led.on
>>> button.when_released = red_led.off
>>> button.when_pressed = buzzer.on
>>> button.when_released = buzzer.off
```

Let's say we want the LED
To light up when we press the button
And off when we release the button

Let's say we want the BUZZER
To buzz when we press the button
And off when we release the button

Exercise 3 – Pedestrian crossing : Algorithm to code



HOW DOES A PEDESTRAIN CROSSING WORK?

RED LED IS ON / GREEN LED IS OFF
WAIT FOR 10 SECONDS
RED LED GOES OFF
GREEN LED GOES ON
GIVE 10 SECONDS FOR PEOPLE TO CROSS
GREEN LED GOES OFF
RED LED TURNS ON

convert

TO CODE

```
from picozero import  LED, Buzzer, Button
from time import sleep
red_led = LED(15)
green_led=LED(14)
buzzer=Buzzer(12)
button=Button(13)
red_led.on()
green_led.off()
sleep(10)
red_led.off()
green_led.on()
sleep(10)
green_led.off()
red_led.on()
```

RUN and
Save this program as ex3c.py

Exercise 3 – Turning the circuit into a pedestrian crossing with code

```
from picozero import  LED, Buzzer, Button
from time import sleep
red_led = LED(15)
green_led=LED(14)
buzzer=Buzzer(12)
button=Button(13)
red_led.on()
green_led.off()
sleep(10)
red_led.off()
green_led.on()
sleep(10)
green_led.off()
red_led.on()
```

RUN ONCE

convert

```
while True:
    red_led.on()
    green_led.off()
    sleep(10)
    red_led.off()
    green_led.on()
    sleep(10)
    green_led.off()
    red_led.on()
```

RUN FOREVER

RUN AND SAVE PROGRAM AS ex3d.py

ex3d.py

```
from picozero import LED, Buzzer, Button
from time import sleep
red_led = LED(15)
green_led=LED(14)
buzzer=Buzzer(12)
button=Button(13)

red_led.on()
green_led.off()

while True:
    sleep(10)
    red_led.off()
    green_led.on()
    sleep(10)
    green_led.off()
    red_led.on()
```

How to convert this into a functon

Give a name for your function. Let's call it greenman()

```
from picozero import LED, Buzzer, Button
from time import sleep
red_led = LED(15)
green_led=LED(14)
buzzer=Buzzer(12)
button=Button(13)
red_led.on()
green_led.off()
def greenman():
    sleep(10)
    red_led.off()
    green_led.on()
    sleep(10)
    green_led.off()
    red_led.on()
while True:
    greenman()
```

Run program
Save it as ex3e.py

ex3e.py

```
from picozero import LED, Buzzer, Button
from time import sleep
red_led = LED(15)
green_led=LED(14)
buzzer=Buzzer(12)
button=Button(13)
red_led.on()
green_led.off()
def greenman():
    sleep(10)
    red_led.off()
    green_led.on()
    sleep(10)
    green_led.off()
    red_led.on()
while True:
    greenman()
```

BEFORE THE GREEN LIGHT GOES OFF, WE WANT TO ALERT THE PEDESTRIAN BY FLASHING THE GREEN LED AND BUZZER



How?
WHERE in the program
to add this function?

```
def countdown():
    for x in range(1,11,1):
        green_led.on()
        buzzer.on()
        sleep(0.5)
        green_led.off()
        buzzer.off()
        sleep(0.5)
```

10 X

ex3e.py

```
from picozero import LED, Buzzer, Button
from time import sleep
red_led=LED(15)
green_led=LED(14)
buzzer=Buzzer(12)
button=Button(13)
red_led.on()
green_led.off()


def greenman():
    sleep(10)
    red_led.off()
    green_led.on()
    sleep(10)
    countdown()
    green_led.off()
    red_led.on()


while True:
    greenman()
```



```
def countdown():
    for x in range(1,11,1):
        green_led.on()
        buzzer.on()
        sleep(0.5)
        green_led.off()
        buzzer.off()
        sleep(0.5)
```

Run program
Save it as ex3f.py

ex3f.py

```
[ ex3f.py ]
1  from picozero import Buzzer, Button, LED
2  from time import sleep
3
4  red_led=LED(15)
5  green_led=LED(14)
6  button=Button(13)
7  buzzer= Buzzer(12)
8
9  red_led.on()
10 green_led.off()
11
12 def countdown():
13     for x in range(1,11,1):
14         green_led.on()
15         buzzer.on()
16         sleep(.5)
17         green_led.off()
18         buzzer.off()
19         sleep(.5)
20
21 def greenman():
22     sleep(10)
23     red_led.off()
24     green_led.on()
25     sleep(10)
26     countdown()
27     green_led.off()
28     red_led.on()
25
26 button.when_pressed = greenman
32
```

This program works fine.
But in the real world, it's not practical.
Why?

If no one wants to cross, the program will still continue to work

We want this program to run only
when someone presses the button. How can we make use of the button

```
26 while True:
27     greenman()
```



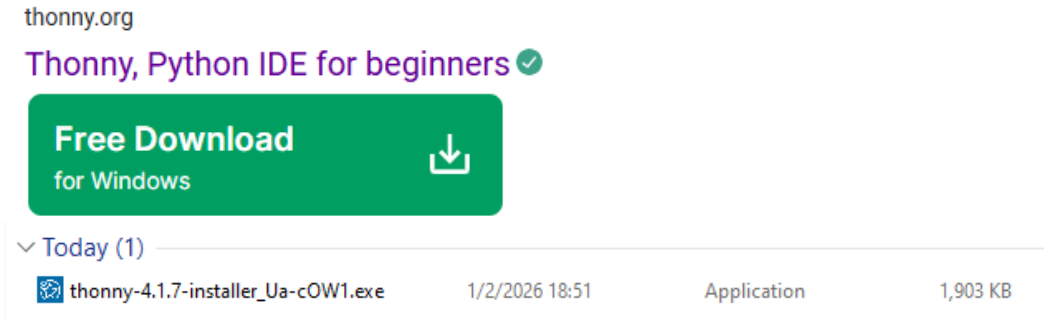
```
25
26 button.when_pressed = greenman
```

Make this change
Run program
Save it as ex3g.py

HOME WORK (OPTIONAL)

Install ChatGPT app on your phone
Bring a thumb drive if you want to save your work

Install Thonny on own laptop/computer:
Search for Thonny in browser



Run the installation. Thonny will be installed together with Python

Follow the lesson on this video
https://youtu.be/A2NnIWCYnMQ?si=gOc3JojI_TgXHxmE

Python Lesson – Session # 3

LIBRARY

- tm1637

FUNCTIONS

FLOW CONTROL

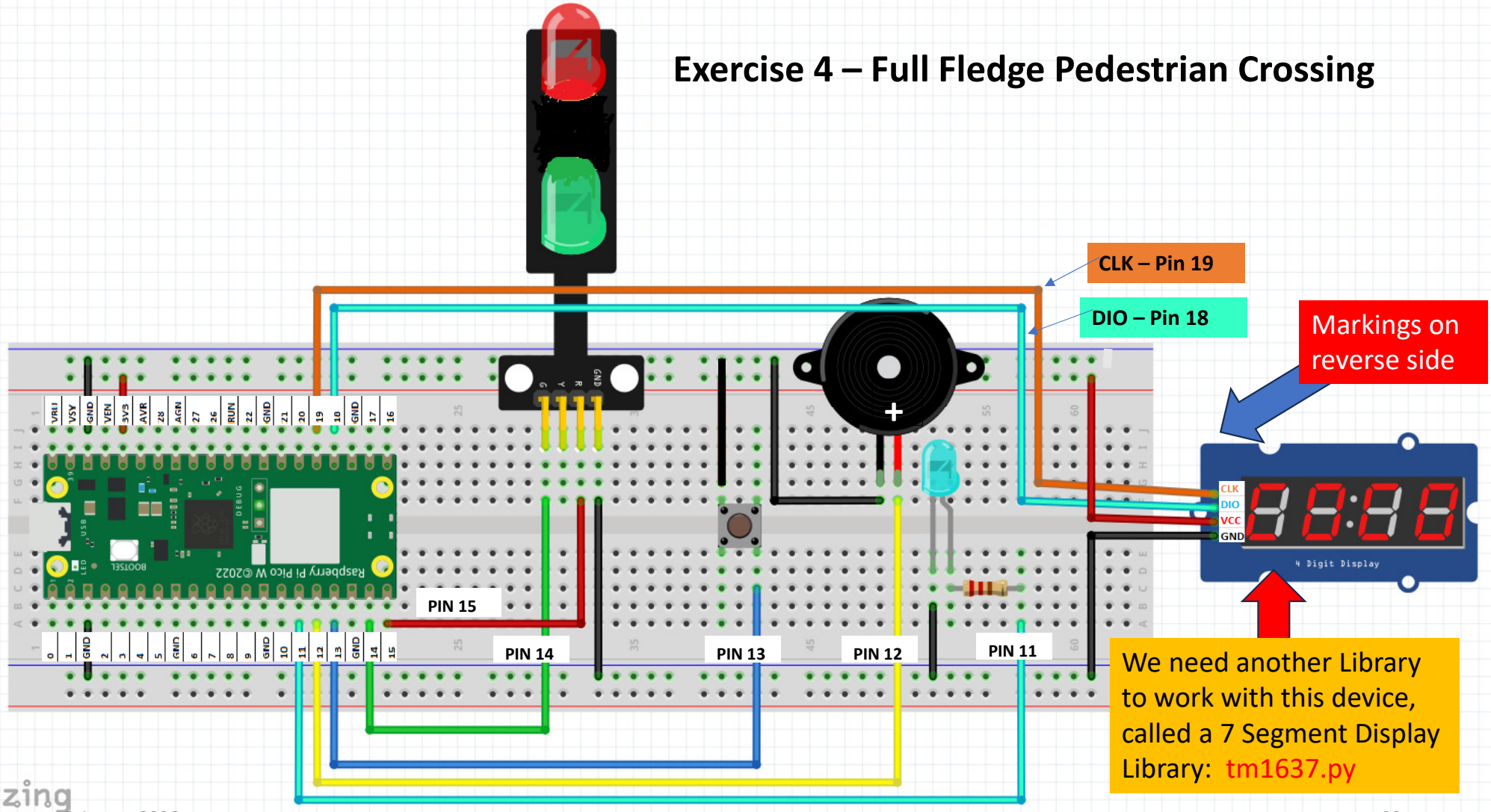
CONDITIONAL STATEMENTS

- **if else**

LOOPS

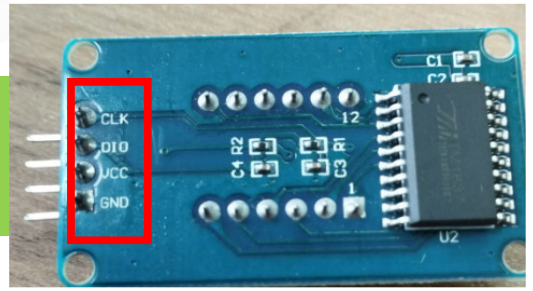
- **for loop**

Exercise 4 – Full Fledge Pedestrian Crossing



EXERCISE 4a – THE 7 SEGMENT DISPLAY






```
>>> from machine import Pin
>>> import tm1637
>>> tm = tm1637.TM1637(clk=Pin(19), dio=Pin(18))
```

```
>>> tm.show("help")
```

```
>>> tm.number(1234)
```

```
>>> tm.temperature(24)
```

```
>>> tm.show(" " * 4)
```

In python " " * 4 means 4 SPACES
This will clear the display.
Make sure there is a SPACE in between the quotation marks

Exercise 4a. Display the following
PICO
95
70 Degrees Celcius
Clear the display

COUNT DOWN DISPLAY

```
from machine import Pin
from time import sleep
import tm1637
tm = tm1637.TM1637(clk=Pin(19), dio=Pin(18))

for x in range(10,-1,-1):
    tm.number(x)
    sleep(1)
```

Run program
Save as ex4b.py

Try Counting from 0 to 10.
Refer to ex3f.py if you are not sure how to count upwards

Modify ex3g.py to add Count down to countdown()

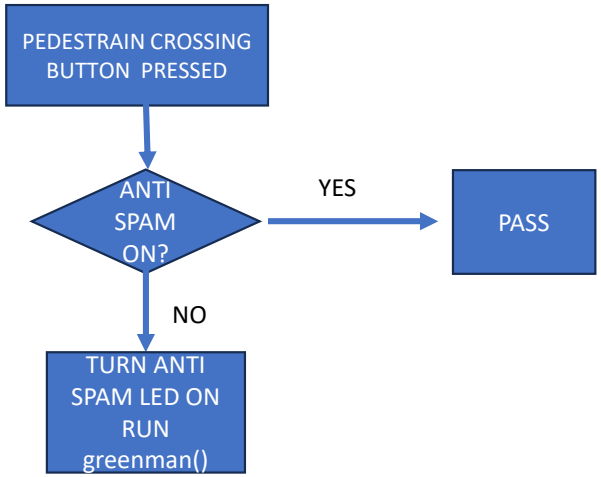
Besides the green man flashing and the buzzer beeping
the counter will show how much time left for crossing countdown()

```
1  from picozero import LED, Buzzer, Button
2  from machine import Pin
3  import tm1637
4  from time import sleep
5  red_led=LED(15)
6  green_led=LED(14)
7  buzzer=Buzzer(12)
8  button=Button(13)
9  tm = tm1637.TM1637(clk=Pin(19), dio=Pin(18))
10 red_led.on()
11 green_led.off()
12 def countdown():
13     for x in range(10,-1,-1):
14         tm.number(x)
15         green_led.on()
16         buzzer.on()
17         sleep(0.5)
18         green_led.off()
19         buzzer.off()
20         sleep(0.5)
21 def greenman():
22     sleep(10)
23     red_led.off()
24     green_led.on()
25     sleep(10)
26     countdown()
27     green_led.off()
28     red_led.on()
29
30 button.when_pressed =    greenman
```

tm.show(" " * 4)

Run this amended program
Save as ex4c.py

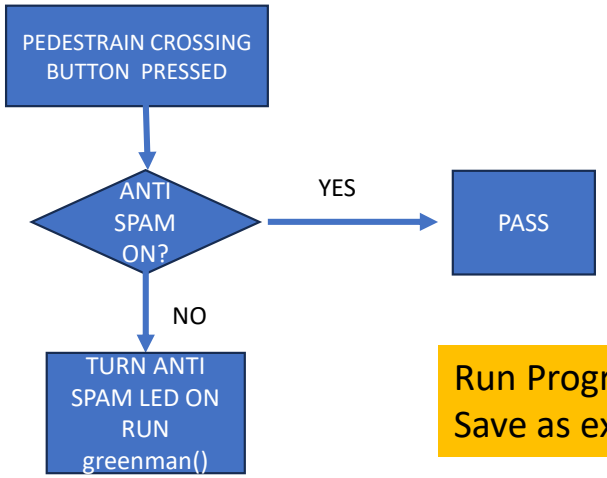
Anti Spamming



MODIFY ex4c.py to add antispam control

```
from picozero import LED, Buzzer, Button
from machine import Pin
import tm1637
from time import sleep
red_led=LED(15)
green_led=LED(14)
buzzer=Buzzer(12)
button=Button(13)
tm=tm1637.TM1637(clk=Pin(19),dio=Pin(18))
antispam_led=LED(11)
antispam_led.off()
red_led.on()
green_led.off()
def countdown():
    for x in range(10,-1,-1):
        tm.number(x)
        green_led.on()
        buzzer.on()
        sleep(0.5)
        green_led.off()
        buzzer.off()
        sleep(0.5)
def check():
    if antispam_led.value == 1:
        pass
    else:
        antispam_led.on()
        greenman()
```

```
def greenman():
    sleep(10)
    red_led.off()
    green_led.on()
    sleep(10)
    countdown()
    green_led.off()
    red_led.on()
    antispam_led.off()
    tm.show(" " * 4)
button.when_pressed = check
```



Run Program
Save as ex4d.py

```
from picozero import LED, Buzzer, Button
from machine import Pin
import tm1637
from time import sleep
red_led=LED(15)
green_led=LED(14)
buzzer=Buzzer(12)
button=Button(13)
tm=tm1637.TM1637(clk=Pin(19),dio=Pin(18))
antispam_led=LED(11)
antispam_led.off()
red_led.on()
green_led.off()
tm.show(" " * 4)
def countdown():
    for x in range(10,-1,-1):
        tm.number(x)
        green_led.on()
        buzzer.on()
        sleep(0.5)
        green_led.off()
        buzzer.off()
        sleep(0.5)
def check():
    if antispam_led.value == 1:
        pass
    else:
        antispam_led.on()
        greenman()
def greenman():
    sleep(10)
    red_led.off()
    green_led.on()
    sleep(10)
    countdown()
    green_led.off()
    red_led.on()
    antispam_led.off()
    tm.show("  " * 4)

button.when_pressed= check
```

Patrol Car Ex – Putting knowledge gained to use

Use your blue , red LED and a Buzzer to make a noisy flashing lights on a patrol car

Choose any pin you wish

Wire up the circuit and write the python code to make it work



HINT

Material

1 x Red LED

1 x Blue LED

1 x Buzzer

Python Code

picozero and time library

while True:

Red LED on and off

Blue LED on and off

Sleep in between on and off

Use this example

```
1  from picozero import LED, Buzzer, Button
2  from time import sleep
3  red_led=LED(15)
4  green_led=LED(14)
5  buzzer=Buzzer(12)
6  button=Button(13)
7  red_led.on()
8  green_led.off()
9
10 def greenman():
11     sleep(10)
12     red_led.off()
13     green_led.on()
14     sleep(10)
15
16     green_led.off()
17     red_led.on()
18
19 while True:
20     greenman()
```

Save your program
As **patrolcar.py**
And RUN

Python Lesson – Session # 4

INTRODUCTION TO CHATGPT

- **GENERATING A PYTHON PROGRAM USING CHATGPT**

Using Prompts

Produce a micropython code using picozero library
To simulate a patrol car with flashing red and blue
Lights and with buzzer as siren

Cancel Send

```
from picozero import LED, Buzzer
from time import sleep

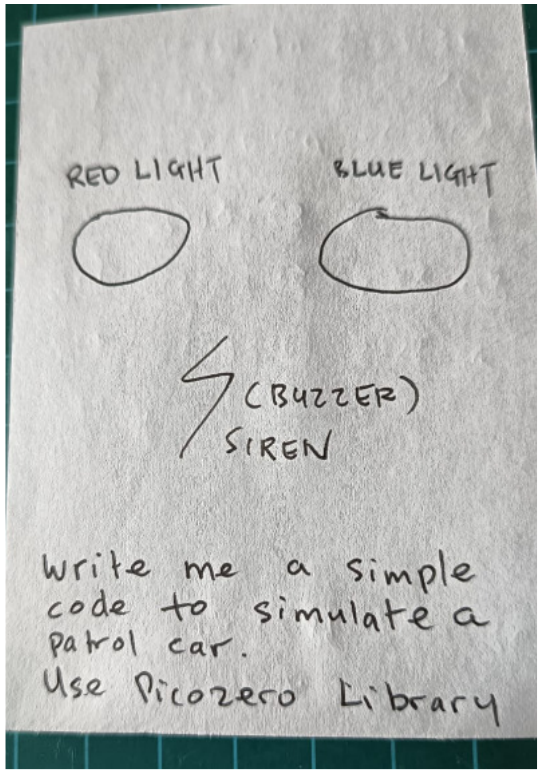
# Devices
red = LED(15)
blue = LED(14)
buzzer = Buzzer(13)

FLASH_TIME = 0.2   # controls speed of lights + siren

while True:
    # RED + SIREN
    red.on()
    blue.off()
    buzzer.on()
    sleep(FLASH_TIME)

    # BLUE + SIREN OFF
    red.off()
    blue.on()
    buzzer.off()
    sleep(FLASH_TIME)
```

Using a sketch



Python Lesson – Session # 4

CONTINUING LEARNING PYTHON ON YOUR OWN
Video Resources

https://github.com/ssgoh/The-Pi-Club/blob/main/Pi%20Python%20Workshop/Recommended%20Video%20Resources.txt



Python Lesson – Session # 4

Sample Training Videos / AI as a Tutor

https://www.youtube.com/watch?v=Ro_MScTDfU4
Learn Python Programming in Only 30 Minutes

- **https://www.youtube.com/watch?v=ix9cRaBkVe0**

Full python course for free

- **https://www.youtube.com/watch?v=sFLRrsIln4k**

How to Learn Python With AI

- **https://www.youtube.com/watch?v=nMCOB8KElwo**

User input in Python is easy + exercises

https://github.com/ssgoh/The-Pi-Club/blob/main/Pi%20Python%20Workshop/Recommended%20Video%20Resources.txt

Python without the Pico

🟢 1. First steps: print and variables

Demo:

Python

```python
print("Hello, world!")
name = "John"
age = 25
print(name, "is", age, "years old")
```

Exercise:

- Change the name and age
- Print: `"Alice is 30 years old"`

Python without the Pico

🟢 2. Input from user

Demo:

Python

```python
name = input("What is your name? ")
print("Hello", name)
```

Exercise:

- Ask for age
- Print: `"Next year you will be X"`

Python without the Pico

🟢 3. Simple math

Demo:

Python

```
a = 10
b = 3
print(a + b)
print(a * b)
```

Exercise:

- Ask user for two numbers
- Print their sum and average

Python without the Pico

🟢 4. If statements (decision making)

Demo:

Python

```python
age = int(input("Enter your age: "))
if age >= 18:
    print("Adult")
else:
    print("Minor")
```

Exercise:

- Check if a number is even or odd

AI Tools as Python Tutor:

ChatGPT
Google AI
Deepseek
Perplexitiy
Gwen

Demo
Will show you how I normally do it
Open IDE and Google AI side by side

DEMO : HOW TO GET CODING ASSISTANCE FROM THE AI

I have the following in my shopping cart
fruits = ["apple", "banana", "orange"]
print(fruits[0])

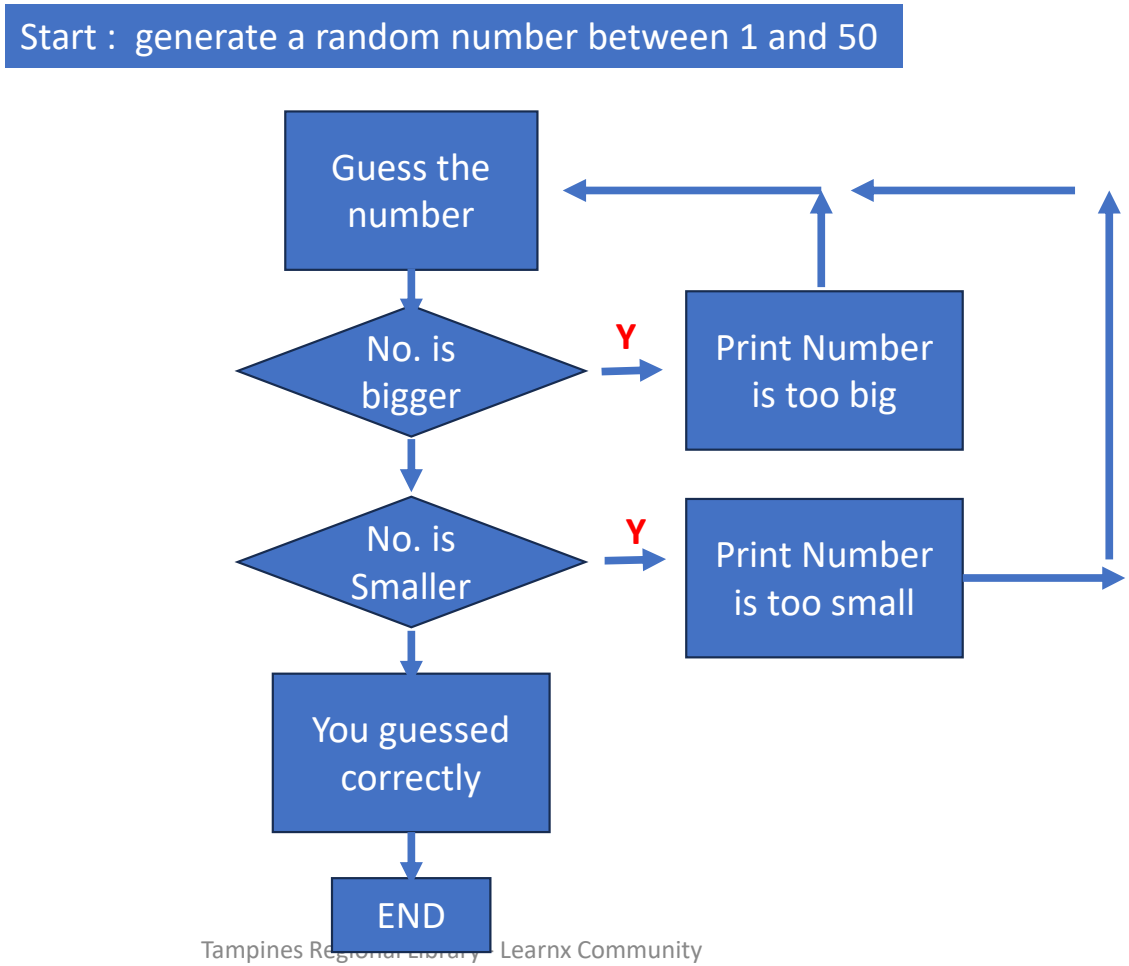
Exercise:
Add a new fruit
Loop through and print all fruits
How to do this?

Explain try: except

```
try:
    a = int(input("Enter a number: "))
    b = int(input("Enter another number: "))
    print("Result =", a / b)

except:
    print("Something went wrong!")
```

Algorithm in the form of a flow chart
for the AI



```
# Create empty lists to hold items and prices
cart_items = []
cart_prices = []

print("=== Welcome to the Shopping Cart ===")

# Loop forever until the user chooses to exit
while True:
    print("\n1. Add Item")
    print("2. View Cart")
    print("3. Checkout")

    choice = input("Enter choice (1, 2, or 3): ")

    if choice == "1":
        # Ask for details and save them directly to the lists
        item = input("Enter item name: ")
        price = float(input("Enter item price: "))

        cart_items.append(item)
        cart_prices.append(price)
        print(item + " added!")

    if choice == "2":
        print("\n--- Items in your Cart ---")
        # Loop through the list using index numbers
        for i in range(len(cart_items)):
            print(cart_items[i] + " - $" + str(cart_prices[i]))

    if choice == "3":
        # Calculate total price and exit the program
        total = sum(cart_prices)
        print("\nFinal Total: $" + str(total))
        print("Goodbye!")
        break
```

Before running the code, make sure you install the library by typing `pip install matplotlib` in your terminal.

```
import matplotlib.pyplot as plt

# 1. Define the data (Simple lists)
days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
steps_walked = [4000, 7500, 6200, 5000, 9000, 11000, 8500]

# 2. Create a basic Line Graph
plt.plot(days, steps_walked, color="green", marker="o")

# 3. Add simple labels and titles
plt.title("My Weekly Steps Tracker")
plt.xlabel("Days of the Week")
plt.ylabel("Number of Steps")

# 4. Display the graph window on your screen
print("Opening the graph... Close the graph window to end the
program.")
plt.show()
```

I am running python on Thonny IDE. Teach me how to set up a text to speech python program